

THE ESG SYNERGY MATRIX

Multi-Site ESG Operational Risk Analytics, Relational Architecture & Decision Support

Target enterprise	Titan Automotive Group - five regional manufacturing hubs (case-study environment)
Technology stack	Microsoft Power BI Desktop - DAX - Power Query (M) - Kimball Star Schema
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Portfolio use	Professional case study for business analytics and BI portfolio presentation

Executive positioning. The ESG Synergy Matrix consolidates production, emissions, energy, safety and renewable-energy indicators into a governed analytical layer. The model is designed to help management identify site-level exceptions, normalize performance metrics and prioritize remediation actions.

Assurance note. The operational figures and contractual assumptions presented in this case study originate from the supplied project model. They should not be interpreted as independently audited corporate data, legal advice or a verified client engagement unless supporting evidence is available.

Prepared for professional portfolio / personal website use
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1. EXECUTIVE SUMMARY

The ESG Synergy Matrix is a Power BI-based decision-support framework designed to integrate production, emissions, energy, safety, renewable-energy usage and compliance attributes across five manufacturing hubs. The analytical design separates transactional measurements from conformed dimensions using a Kimball-style star schema, enabling consistent aggregation and site-level comparison.

The supplied case-study dataset contains 178,800 units of output, 7,646.6 metric tonnes of gross CO2, \$685,500 in energy cost and 7 safety incidents across the five hubs. These figures are reproduced from the project source and are presented as analytical case data rather than audited corporate disclosures.

178,800 Units standardized	7,646.6 MT Gross CO2 in case data	\$685.5K Energy cost in case data	5 Manufacturing hubs
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Key analytical finding

Operational scale and operational risk are not equivalent variables. The model therefore uses normalized rates and threshold-based measures rather than relying solely on absolute incident counts or production volume.

Portfolio interpretation

The project is positioned as a technical business-analysis case study demonstrating data modeling, KPI design, DAX logic, exception analysis and management decision support.

2. BUSINESS PROBLEM

The case assumes a five-site manufacturing network where operational and ESG information is fragmented across localized records. The business-analysis objective is to create a common analytical layer that allows management to compare sites, identify exceptions, normalize risk indicators and support remediation workflows.

Business questions supported by the model

- Which facilities show elevated safety or environmental indicators after adjusting for production volume?
- Where do energy cost, emissions, renewable-energy usage and production output diverge?
- Which sites require management attention under approved contractual or internal thresholds?
- Which operating practices can be transferred from lower-incident sites to locations requiring remediation?

3. DATA ARCHITECTURE

The model uses a Kimball-style star schema. FACT_ESG stores atomic operational observations, while conformed dimensions provide facility, calendar and compliance context. This structure supports controlled filter propagation and reduces duplicated aggregation during analysis.

Layer	Object	Purpose
Fact	FACT_ESG	DateKey; LocationKey; Units_Produced; Scope 1 / Scope 2 CO2; Energy Cost; Safety Incidents; Renewable %
Dimension	Dim_Facility	Facility attributes and regional context
Dimension	Dim_Calendar	Date hierarchy and reporting periods
Dimension	Dim_ComplianceTier	Approved threshold and compliance classification logic

4. ANALYTICAL INTELLIGENCE - DAX

The model normalizes safety incidents against production volume so that facilities with different throughput can be compared using a common rate.

Normalized Safety Incident Rate - per 1,000 units

Safety Rate = DIVIDE(SUM('FACT_ESG'[Safety_Incidents]), SUM('FACT_ESG'[Units_Produced]), 0) * 1000

Contractual Risk Measure - source model

Contractual Risk (\$) = VAR IncidentCeiling = 2 VAR PenaltyPerBreach = 2500000 VAR ActualIncidents = SUM('FACT_ESG'[Safety_Incidents]) RETURN IF(ActualIncidents > IncidentCeiling, (ActualIncidents - IncidentCeiling) * PenaltyPerBreach, 0)

Control check. Under the stated formula, 3 incidents with a ceiling of 2 and a \$2.5M incremental penalty produce \$2.5M of calculated exposure. Any wider portfolio figure requires a separately defined aggregation rule and supporting contractual evidence.

5. OPERATIONAL AUDIT - SUPPLIED CASE DATA

Hub	Output	Gross CO2	Energy Cost	Incidents	Renewable	Status
Chennai	42,700	1,802.8 MT	\$160,000	3	12.3%	Threshold exception
Delhi	38,300	1,661.5 MT	\$153,000	1	17.3%	Watch
Gurgaon	36,300	1,597.3 MT	\$144,500	2	21.0%	Moderate
Mumbai	34,500	1,523.0 MT	\$138,000	1	25.7%	Controlled
Pune	27,000	1,062.0 MT	\$90,000	0	51.7%	Reference site
TOTAL	178,800	7,646.6 MT	\$685,500	7	28.4% avg	Portfolio view

The operational values above are transcribed from the supplied project source. They should be treated as portfolio-case data until the underlying source files, contractual terms, calculation logic and measurement methodology are independently verified.

6. MANAGEMENT ACTION ROADMAP

Phase	Action	Decision logic
1 - 30 days	Investigate Chennai exception	Validate incident classification, contractual thresholds, emissions data and root cause before committing capital.
2 - 60 days	Transfer proven safety practices	Document the reference site's process and test applicability at higher-risk locations.
3 - Continuous	Automated threshold monitoring	Deploy Power BI alerts or dashboard controls against approved internal or contractual thresholds.

7. REGULATORY & GOVERNANCE CONTEXT

The analytical model is designed to support ESG reporting and operational governance across jurisdictions. Regulatory applicability should be assessed for the specific entity, reporting perimeter, industry, contractual obligations and operating countries. Reference frameworks may include IFRS Sustainability Disclosure Standards, the EU Corporate Sustainability Due Diligence framework and applicable U.S. securities-disclosure requirements.

Governance controls recommended for production deployment

- Document source-system ownership, refresh frequency, data lineage and transformation rules.
- Maintain controlled definitions for every KPI, threshold, denominator and reporting period.
- Separate contractual thresholds from regulatory thresholds; they are not interchangeable.
- Version-control DAX measures and business rules, including edge-case and denominator testing.
- Retain evidence supporting any financial-exposure calculation before presenting it as a liability or contractual claim.
- Use role-based access and a documented change-control process for enterprise deployment.

8. LIMITATIONS & ASSURANCE

This portfolio case study demonstrates analytical design and business-analysis reasoning. It does not constitute legal advice, an ESG assurance statement, an independent audit or proof of a live commercial engagement. The operational dataset, contractual penalty assumptions and financial-exposure figures were supplied as part of the project model and require primary-source validation before operational use.

Decision-use principle. Use the dashboard to identify exceptions and support management investigation; do not treat model output as a substitute for contract review, statutory reporting controls, internal audit or independent assurance.

9. PORTFOLIO POSITIONING

Recommended website description: "A Power BI ESG operations case study demonstrating star-schema data modeling, DAX-based normalization, site-level exception analysis and management decision support across a five-facility manufacturing network."

The project demonstrates technical business-analysis capability while keeping the distinction between analytical output and verified corporate evidence explicit.

APPENDIX A - QUALITY & GOVERNANCE CHECKLIST

Area	Current standard	Portfolio treatment
Document design	Professional	Consistent typography, tables, page numbering and restrained visual hierarchy
Business narrative	Case-study appropriate	Claims framed as analytical outputs, assumptions and management implications
DAX consistency	Defined	Formula logic reconciled with displayed incident-based exposure
Regulatory context	Jurisdiction-specific	No blanket statement that one regime applies universally
Audit language	Controlled	No use of “audit-proof”, “certified” or equivalent assurance claims without evidence
Data provenance	Required	Source lineage should be added for any production implementation
Financial exposure	Subject to validation	Contract terms and calculation methodology must be evidenced before executive use

METHODOLOGY & REFERENCE NOTES

Primary project source: TITAN-ERM-2026-CP | Executive Technical White Paper. The project source provides the Power BI, DAX, Power Query and Kimball architecture and the five-site operational figures reproduced in this case study.

Reference frameworks: IFRS Sustainability Disclosure Standards (IFRS S1 / S2); EU Corporate Sustainability Due Diligence framework; applicable U.S. securities-disclosure requirements. Regulatory applicability must be determined for the specific entity and jurisdiction.

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